MIL-M-36582A (DSA-DM)
5 June 1975
SUPERSEDING
MIL-M-36582 (DSA-DM)
10 November 1970

MILITARY SPECIFICATION

MICROTOME, ROTARY, AUTOMATIC, REFRIGERATED

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been reviewed for promulgation as a coordinated revision of MIL-M-36582 (DSA-DM). It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of refrigerated microtome suitable for use in freezing and cutting frozen tissue.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640	Box, Fiberboard, Corrugated, Triple-wall.

Military

MIL-K-36311	Knives, Microtome.
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-M-7298	Manual, Technical, Commercial Equipment.

FSC 6640

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STANDARDS

Federal

FED-STD-66	Steel, Chemical Composition and Hardenability.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (A.N.S.I.)

A.N.S.I.	C73-11	Dimensions of Plugs and Receptacles - General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3-Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

UNDERWRITERS' LABORATORIES, INC.

UL 471	Commercial Refrigerators

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., N.Y. 11746; 207 East Ohio Street, Chicago, Ill. 60611; and 1655 Scott Boulevard, Santa Clara, Calif. 95050.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout and as herein specified.

3.1.1 Corrosion-resisting steel. Corrosion-resisting steel shall be in accordance with FED-STD-66, 300 Series.

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3.2 Fire and casualty hazards. The contractor must submit proof prior to delivery that the items to be delivered under the contract conform to the applicable requirements of Underwriters' Laboratories, Inc. UL 471. Proof shall consist of one of the following:

a. UL label, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, upon the latter's request.

b. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.3 Construction. The refrigerated microtome shall consist of a microtome and a refrigeration unit both enclosed in a cabinet. Maximum overall dimensions in inches shall be: height 50, width 30 and depth 32.

* 3.3.1 Cabinet. The cabinet shall be fabricated of steel or aluminum. The lower portion of the cabinet shall house the refrigeration unit, and the upper portion shall be a cold chamber which shall house the microtome. The four bottom corners shall have casters for mobility; two of the casters shall have locking brakes. The top of the cabinet shall have a transparent cover. Cover shall be of the sliding or hinged type. Top or right hand side of cabinet shall have a control panel. Refrigerated microtome shall be so constructed that a seated operator can view the interior of the microtome housing and the controls on the control panel from one position. Lower portion of cabinet shall have removable panels which shall be so placed as to provide openings for servicing of the refrigeration unit. For purposes of this specification, the portion of the refrigerated microtome directly in front of the operator shall be defined as the front of the refrigerated microtome.

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3.3.2 Refrigeration unit. Refrigeration unit shall consist of a compressor and a motor. Refrigeration unit shall be hermetically sealed and shall be mounted under the cold chamber. Motor shall produce a minimum of 1/3 horsepower, shall be suitable for use with 115 volt (nominal), 60 hz alternating current and shall be supplied with a cord and plug.

3.3.2.1 Cord and plug. Cord shall be a three-wire power cord at least 6 feet long. A plug in accordance with ANSI C73.11 shall be secured to the end of the cord.

* 3.3.3 Microtome housing (cold chamber). The sides and bottom of the housing shall be fabricated of corrosion-resisting sheet steel. Housing shall be of one-piece construction or of multipiece construction. If multipiece construction is used, sheets shall be welded together with welding rods of the same composition as the sheet steel. The bottom shall have a hole which connects to a drainage pipe. Drainage pipe shall extend to the outside of the cabinet. The right side of the housing (when viewed from the front) shall have means such as a hole or a retractable bearing for connection of handwheel to the microtome by insertion of the handwheel shaft. A shelf shall be secured to one of the sides. Shelf shall be fabricated of copper or of the same material as the sides and bottom and shall be capable of holding four specimen holders.

* 3.3.4 Control panel. Control panel shall contain an off-on switch, a temperature-indicating scale and a temperature control device. Temperature-indicating scale shall be graduated from - 30 degrees Centigrade (°C.) to 0°C. Minimum interval shall not exceed 2°C. -30°C., - 20°C., -10°C. shall be marked with applicable Arabic numbers.

3.3.5 Microtome. The microtome shall be a durable, accurate, automatic feed, direct-drive, rotary type and shall be constructed of corrosion-resisting metals. The microtome shall be capable of cutting unwrinkled frozen tissue sections in thicknesses ranging from 2 to 30 microns. The microtome shall consist of a knife holder and feed mechanism.

3.3.5.1 Knife holder. Knife holder shall be fabricated of corrosion-resisting steel and shall be capable of providing rigid support for the microtome knife. Means shall be provided for lateral and tilt adjustments of the knife.

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3.3.5.2 Feed mechanism. Feed mechanism shall be activated by a handwheel which can be mounted on the outside of the cabinet in the position indicated in 3.3.3. Feed mechanism shall move the knife holder or the specimen holder clamp, so that the knife holder and specimen clamp approach each other when the handwheel is turned clockwise. The amount of movement shall be determined by a conveniently placed control. Feed movement shall be coordinated with cutting action. Cutting action shall be caused by vertical movement of the specimen clamp. Control, feed mechanism and vertical movement shall be so synchronized that the feed mechanism moves forward the amount marked on control during one vertical excursion of the specimen clamp. Control shall be marked from 2 to 16 in increments of not more than 2.

3.3.5.2.1 Specimen holder clamp. Clamp shall be provided for securing and orienting specimen holder.

3.4 Performance.

3.4.1 Temperature depression of knife blade. When tested as specified in 4.4.1, the knife blade shall drop from 23°C. to -25°C. within 6 hours, and the temperature reading on the control panel and knife blade shall be identical $\pm$ 2°C.

3.4.2 Feed mechanism. The feed mechanism shall move the number of microns set on the thickness control $\pm$ 1 micron when tested as specified in 4.4.2.

3.4.3 Corrosion resistance. The microtome, specimen holder and microtome housing shall show no evidence of corrosion when tested as specified in 4.4.3.

3.4.4 Vibration. The cabinet shall not oscillate more than 1/4 inch when tested as specified in 4.4.4.

3.4.5 Dielectric strength. There shall be no breakdown when the refrigerated microtome is subjected to the dielectric strength test specified in 4.4.5.

* 3.4.6 Operational capability. When tested as specified in 4.4.6, temperature in cold chamber shall drop to -25° C. When defrosted, ice, frost and moisture shall be removed from the interior of the chamber and cover. Noise level shall not exceed the noise level of the average home refrigerator while compressor is running.

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3.5 Finish. Surfaces shall be corrosion-resisting. If carbon steel is used, surface shall be plated or enameled. Before enameling carbon steel, surfaces shall be cleaned and degreased, treated with zinc phosphate and covered with a coat of primer.

3.6 Service Data. The contractor shall furnish, with each unit, three copies each of a technical manual which contains complete instructions for installation, operation, defrosting, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, three preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the idential item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.7 Accessories. The microtome shall be supplied with one each of the following accessories: knife, specimen holder, anti-roll device and a container of low temperature oil. At least 1 ounce of oil shall be furnished.

3.7.1 Knife. Knife shall be in accordance with MIL-K-36311, size 1 or size 2.

3.7.2 Specimen holder. Specimen holder shall be fabricated from corrosion-resisting metal. The holder shall be so designed that it holds the specimen in a cutting position while the specimen holder mates with and fits into the specimen clamp.

3.7.3 Anti-roll device. Anti-roll device shall be so designed that it can be secured on or near the knife for the prevention of curling of specimens.

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3.8 Identification marking. Each refrigerated microtome shall be marked with the following information:

a. Contract number
b. Name or registered trademark of the manufacturer and model number
c. Electrical data

Marking shall be legible and permanent and shall be accomplished by stamping or by the use of metal plates permanently affixed. Markings secured by adhesives shall not be used.

3.9 Workmanship. Workmanship shall be first class throughout. Refrigerated microtome shall be so constructed that microtome can be easily inserted and removed by following instructions specified in 3.6. Refrigerated microtomes shall be free of defects listed in Table III and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturers of the knives and corrosion-resisting steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

Table I. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major A defects	II	1.0
Major B defects	S-1	1.0
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Acceptance number for all tests shall be zero.

* Table II. Sampling for tests

Component	Characteristic	Requirement	Test Procedure	Inspection level
Knife blade	Temperature depression	3.4.1	4.4.1	S-2
Feed mechanism	Sample thickness	3.4.2	4.4.2	S-2
Microtome, specimen holder and microtome housing	Corrosion resistance	3.4.3	4.4.3	S-2
Refrigerated microtome	Vibration	3.4.4	4.4.4	S-2

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Table II. Sampling for tests (cont'd)

Component	Characteristic	Requirement	Test Procedure	Inspection level
Refrigerated microtome	Dielectric strength	3.4.5	4.4.5	*
Refrigerated microtome	Operation	3.4.6	4.4.6	*

* 100% of refrigerated microtomes shall be tested.

4.3 Examination. The refrigerated microtomes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification defects. Examination shall be conducted in accordance with classification of defects but shall not be limited to the defects as indicated in Table III.

Table III Classification of defects

Category	Defect
Major A	
101	Component missing.
102	Service data missing.
103	Unit not free of sharp edges (other than knives).
104	Unit not free of buckling or dents.
105	Unit not free of corroded areas.
106	Applicable parts or components not secured.
Major B	
151	Microtome not easily inserted or removed by following instructions in service data.
Minor	
201	Finish not free of blisters or cracks.
202	Finish not continuous.
203	Unit not free of scratches.
204	Identification marking not complete, correct, legible or permanent.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.4.1 Temperature depression of knife blade. The refrigerated microtome with knife in place shall be conditioned at an ambient temperature of 23 ± 2°C. for at least 24 hours. After conditioning, the cover shall be closed and the temperature control device shall be so positioned that the temperature of the knife edge will reach -25°C. Temperature-sensing device of known accurancy shall be placed on the knife edge before the cover is closed and shall be so designed and positioned that it measures the temperature of the knife edge.

4.4.2 Feed mechanism. A gauge which is capable of measuring the distance between the knife holder and the specimen clamp or specimen holder shall be secured to the knife holder or specimen clamp. Thickness control shall be set at 10 microns. Microtome shall be operated in accordance with supplier's instructions (see 3.6). Specimen holder may be inserted in specimen clamp. Test gauge shall be of known accuracy. Any tolerance on test gauge shall be subtracted from the tolerance specified in 3.4.2.

* 4.4.3 Corrosion resistance. Microtome, specimen holder and microtome housing shall be exposed to 85% humidity (minimum) at 25°C. ± 2° for 3 hours.

4.4.4 Vibration test. Stationary measuring means shall be placed around all four edges in such a manner that movement in excess of requirements of 3.4.4 will be detected. Caster brakes shall be locked and refrigeration unit turned on and allowed to run for five minutes.

4.4.5 Dielectric strength test. A 60-cycle alternating potential of 900 volts shall be applied for a period of one minute between the current-carrying parts and the dead metal parts. The test shall be conducted with a suitable transformer which can regulate the output voltage. Starting at zero, the applied potential shall be increased gradually until 900 volts is reached or breakdown occurs.

* 4.4.6 Operational test. Refrigeration unit shall be activated and shall be allowed to run until the temperature in the cold chamber is depressed to -25°C. After temperature has reached -25°C., cold chamber shall be defrosted in accordance with supplier's instructions (see 3.6). Indicator for temperature-measuring device shall be so positioned that it can be read easily, and temperature-measuring device shall be so calibrated that the tolerance on accuracy indications at -25°C. is not more than ± 2°C.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each microtome shall be disassembled as far as is practicable including removal of the casters. Component parts, with accessories, shall be packaged within the cold chamber of the unit in such a manner as to insure against shifting of or damage to the unit. The slide block of the microtome shall be immobilized at the top of its travel by a fiberboard block or other suitable means. The unit shall be further packaged in an upright position in a fiberboard box of appropriate size and design. The box shall be designed with suitable interior fittings and cushioning material as required to prevent shifting, marring or other damage to the contents.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B, or C (see 6.2).

* 5.2.1 Level A. Each microtome, packaged as specified in 5.1.1.1, shall be overboxed in an exterior container designed for a type 3 load and constructed in accordance with PPP-B-601, overseas type or PPP-B-621, class 2. Exterior container design shall include two skids of not less than 1-1/2 inches in thickness applied crosswise of the base so positioned in an upright position. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraghs of appendix thereto.

5.2.2 Level B. Each microtome, packed as specified in 5.1.1.1, shall be overboxed in an exterior container designed for a type 3 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; or PPP-B-640, class 2. Closure shall be as specified in the appendix of the applicable box specification. The exterior container shall include two skids of not less than 1-1/2 inches in thickness applied crosswise of the base, positioned so as to facilitate handling with a forklift truck and to maintain the unit in an upright position.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Interior packages shall be marked to identify contents.

* 5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, special handling markings as specified in MIL-STD-129 shall also be shown.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. National Stock Number (NSN).

6.2 This specification covers the following item as listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-982-8223	MICROTOME, Rotary, Automatic, Refrigerated

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0929
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-36582A (DSA-DM), MICROTOME, ROTARY, AUTOMATIC, REFRIGERATED

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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